

GEOMETRY

Friday, June 21, 2019 — 9:15 a.m. to 12:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 35 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator, a straightedge (ruler), and a compass must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

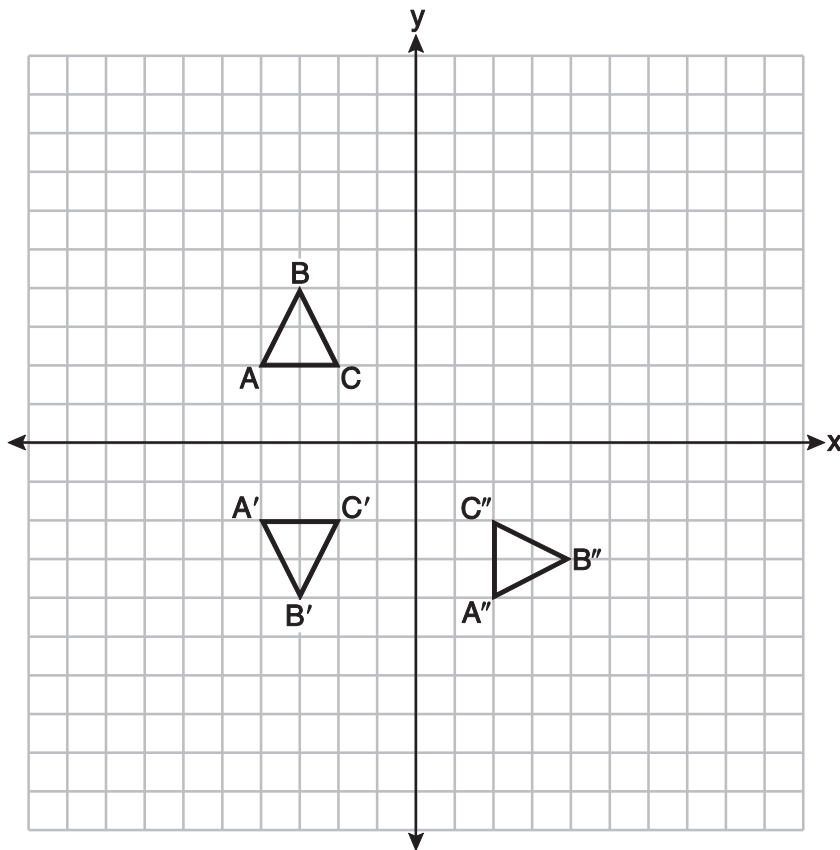
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 On the set of axes below, triangle ABC is graphed. Triangles $A'B'C'$ and $A''B''C''$, the images of triangle ABC , are graphed after a sequence of rigid motions.



Identify which sequence of rigid motions maps $\triangle ABC$ onto $\triangle A'B'C'$ and then maps $\triangle A'B'C'$ onto $\triangle A''B''C''$.

- (1) a rotation followed by another rotation
- (2) a translation followed by a reflection
- (3) a reflection followed by a translation
- (4) a reflection followed by a rotation

**Use this space for
computations.**

- 2** The table below shows the population and land area, in square miles, of four counties in New York State at the turn of the century.

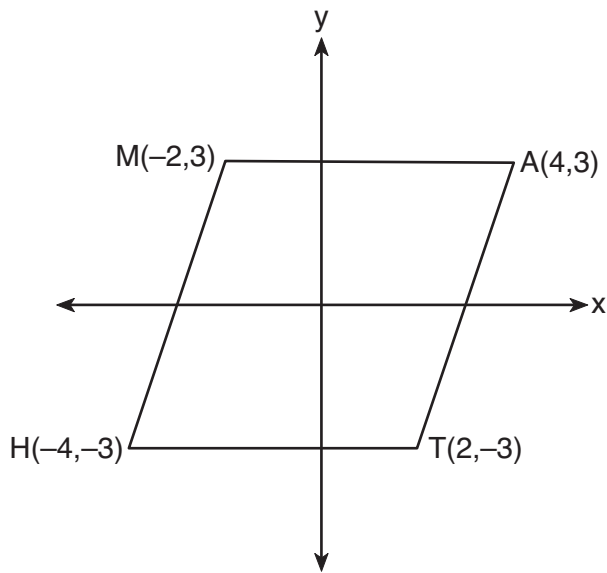
County	2000 Census Population	2000 Land Area (mi²)
Broome	200,536	706.82
Dutchess	280,150	801.59
Niagara	219,846	522.95
Saratoga	200,635	811.84

Which county had the greatest population density?

- (1) Broome (3) Niagara
(2) Dutchess (4) Saratoga
- 3** If a rectangle is continuously rotated around one of its sides, what is the three-dimensional figure formed?
- (1) rectangular prism (3) sphere
(2) cylinder (4) cone

**Use this space for
computations.**

4 Which transformation carries the parallelogram below onto itself?



- (1) a reflection over $y = x$
- (2) a reflection over $y = -x$
- (3) a rotation of 90° counterclockwise about the origin
- (4) a rotation of 180° counterclockwise about the origin

5 After a dilation centered at the origin, the image of $\overline{CD}$ is $\overline{C'D'}$. If the coordinates of the endpoints of these segments are $C(6, -4)$, $D(2, -8)$, $C'(9, -6)$, and $D'(3, -12)$, the scale factor of the dilation is

- (1) $\frac{3}{2}$
- (2) $\frac{2}{3}$
- (3) 3
- (4) $\frac{1}{3}$

**Use this space for
computations.**

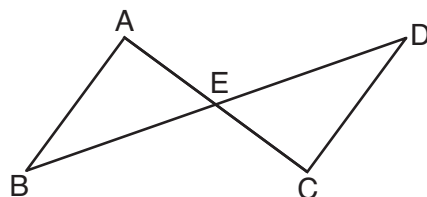
- 6** A tent is in the shape of a right pyramid with a square floor. The square floor has side lengths of 8 feet. If the height of the tent at its center is 6 feet, what is the volume of the tent, in cubic feet?

(1) 48 (3) 192
(2) 128 (4) 384

- 7** The line $-3x + 4y = 8$ is transformed by a dilation centered at the origin. Which linear equation could represent its image?

(1) $y = \frac{4}{3}x + 8$ (3) $y = -\frac{3}{4}x - 8$
(2) $y = \frac{3}{4}x + 8$ (4) $y = -\frac{4}{3}x - 8$

- 8** In the diagram below, $\overline{AC}$ and $\overline{BD}$ intersect at E .



Which information is always sufficient to prove $\triangle ABE \cong \triangle CDE$?

- (1) $\overline{AB} \parallel \overline{CD}$
(2) $\overline{AB} \cong \overline{CD}$ and $\overline{BE} \cong \overline{DE}$
(3) E is the midpoint of $\overline{AC}$.
(4) $\overline{BD}$ and $\overline{AC}$ bisect each other.

- 9** The expression $\sin 57^\circ$ is equal to

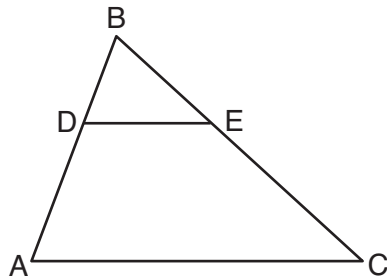
(1) $\tan 33^\circ$ (3) $\tan 57^\circ$
(2) $\cos 33^\circ$ (4) $\cos 57^\circ$

**Use this space for
computations.**

10 What is the volume of a hemisphere that has a diameter of 12.6 cm, to the *nearest tenth of a cubic centimeter*?

- (1) 523.7 (3) 4189.6
(2) 1047.4 (4) 8379.2

11 In the diagram below of $\triangle ABC$, D is a point on $\overline{BA}$, E is a point on $\overline{BC}$, and $\overline{DE}$ is drawn.



If $BD = 5$, $DA = 12$, and $BE = 7$, what is the length of $\overline{BC}$ so that $\overline{AC} \parallel \overline{DE}$?

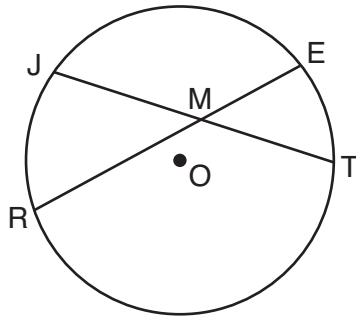
- (1) 23.8 (3) 15.6
(2) 16.8 (4) 8.6

12 A quadrilateral must be a parallelogram if

- (1) one pair of sides is parallel and one pair of angles is congruent
(2) one pair of sides is congruent and one pair of angles is congruent
(3) one pair of sides is both parallel and congruent
(4) the diagonals are congruent

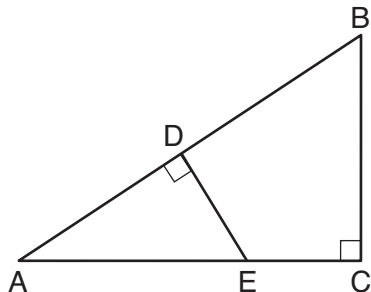
Use this space for
computations.

- 13 In the diagram below of circle O , chords $\overline{JT}$ and $\overline{ER}$ intersect at M .



If $EM = 8$ and $RM = 15$, the lengths of $\overline{JM}$ and $\overline{TM}$ could be

- (1) 12 and 9.5 (3) 16 and 7.5
(2) 14 and 8.5 (4) 18 and 6.5
- 14 Triangles $\triangle JOE$ and $\triangle SAM$ are drawn such that $\angle E \cong \angle M$ and $\overline{EJ} \cong \overline{MS}$. Which mapping would *not* always lead to $\triangle JOE \cong \triangle SAM$?
- (1) $\angle J$ maps onto $\angle S$ (3) $\overline{EO}$ maps onto $\overline{MA}$
(2) $\angle O$ maps onto $\angle A$ (4) $\overline{JO}$ maps onto $\overline{SA}$
- 15 In $\triangle ABC$ shown below, $\angle ACB$ is a right angle, E is a point on $\overline{AC}$, and $\overline{ED}$ is drawn perpendicular to hypotenuse $\overline{AB}$.



If $AB = 9$, $BC = 6$, and $DE = 4$, what is the length of $\overline{AE}$?

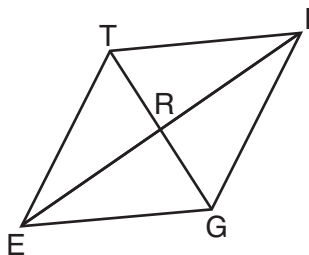
- (1) 5 (3) 7
(2) 6 (4) 8

Use this space for
computations.

- 16 Which equation represents a line parallel to the line whose equation is $-2x + 3y = -4$ and passes through the point $(1,3)$?

- (1) $y - 3 = -\frac{3}{2}(x - 1)$ (3) $y + 3 = -\frac{3}{2}(x + 1)$
(2) $y - 3 = \frac{2}{3}(x - 1)$ (4) $y + 3 = \frac{2}{3}(x + 1)$

- 17 In rhombus $TIGE$, diagonals $\overline{TG}$ and $\overline{IE}$ intersect at R . The perimeter of $TIGE$ is 68, and $TG = 16$.



What is the length of diagonal $\overline{IE}$?

- (1) 15 (3) 34
(2) 30 (4) 52
- 18 In circle O two secants, $\overline{ABP}$ and $\overline{CDP}$, are drawn to external point P . If $m\widehat{AC} = 72^\circ$, and $m\widehat{BD} = 34^\circ$, what is the measure of $\angle P$?
- (1) 19° (3) 53°
(2) 38° (4) 106°

**Use this space for
computations.**

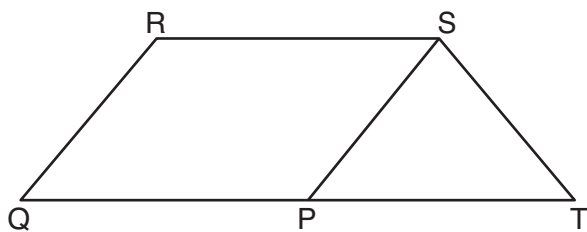
- 19** What are the coordinates of point C on the directed segment from $A(-8,4)$ to $B(10,-2)$ that partitions the segment such that $AC:CB$ is 2:1?

- (1) $(1,1)$ (3) $(2,-2)$
(2) $(-2,2)$ (4) $(4,0)$

- 20** The equation of a circle is $x^2 + 8x + y^2 - 12y = 144$. What are the coordinates of the center and the length of the radius of the circle?

- (1) center $(4,-6)$ and radius 12
(2) center $(-4,6)$ and radius 12
(3) center $(4,-6)$ and radius 14
(4) center $(-4,6)$ and radius 14

- 21** In parallelogram $PQRS$, $\overline{QP}$ is extended to point T and $\overline{ST}$ is drawn.



If $\overline{ST} \cong \overline{SP}$ and $m\angle R = 130^\circ$, what is $m\angle PST$?

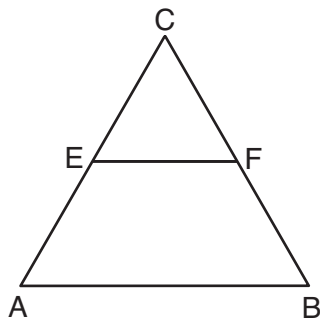
- (1) 130° (3) 65°
(2) 80° (4) 50°

**Use this space for
computations.**

- 22** A 12-foot ladder leans against a building and reaches a window 10 feet above ground. What is the measure of the angle, to the *nearest degree*, that the ladder forms with the ground?

- (1) 34 (3) 50
(2) 40 (4) 56

- 23** In the diagram of equilateral triangle ABC shown below, E and F are the midpoints of $\overline{AC}$ and $\overline{BC}$, respectively.



If $EF = 2x + 8$ and $AB = 7x - 2$, what is the perimeter of trapezoid $ABFE$?

- (1) 36 (3) 100
(2) 60 (4) 120

- 24** Which information is *not* sufficient to prove that a parallelogram is a square?

- (1) The diagonals are both congruent and perpendicular.
(2) The diagonals are congruent and one pair of adjacent sides are congruent.
(3) The diagonals are perpendicular and one pair of adjacent sides are congruent.
(4) The diagonals are perpendicular and one pair of adjacent sides are perpendicular.
-

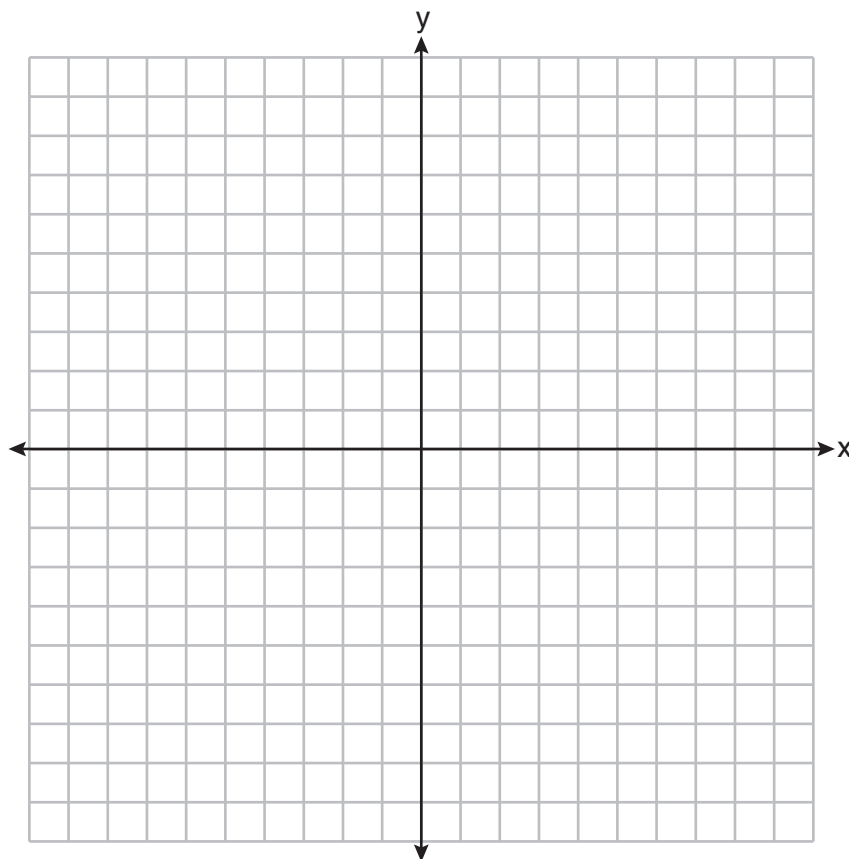
Part II

Answer all 7 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [14]

- 25 Triangle $A'B'C'$ is the image of triangle ABC after a dilation with a scale factor of $\frac{1}{2}$ and centered at point A . Is triangle ABC congruent to triangle $A'B'C'$? Explain your answer.

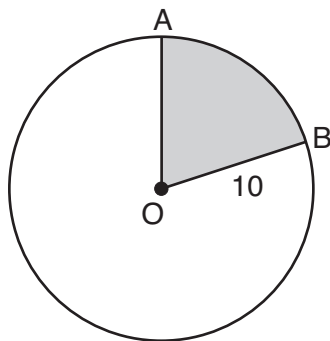
26 Determine and state the area of triangle PQR , whose vertices have coordinates $P(-2, -5)$, $Q(3, 5)$, and $R(6, 1)$.

[The use of the set of axes below is optional.]



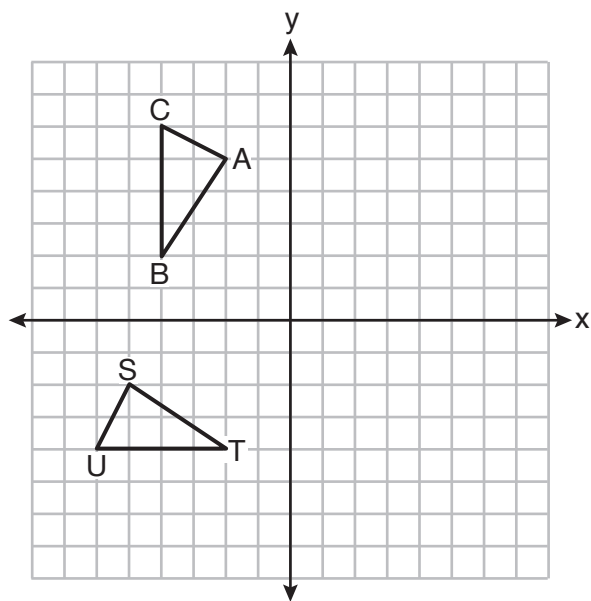
- 27** A support wire reaches from the top of a pole to a clamp on the ground. The pole is perpendicular to the level ground and the clamp is 10 feet from the base of the pole. The support wire makes a 68° angle with the ground. Find the length of the support wire to the *nearest foot*.

28 In the diagram below, circle O has a radius of 10.



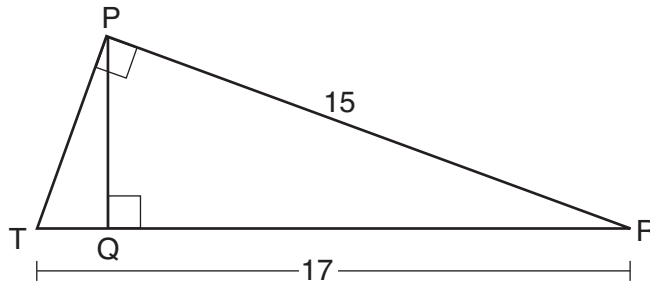
If $m\widehat{AB} = 72^\circ$, find the area of shaded sector AOB , in terms of π .

29 On the set of axes below, $\triangle ABC \cong \triangle STU$.



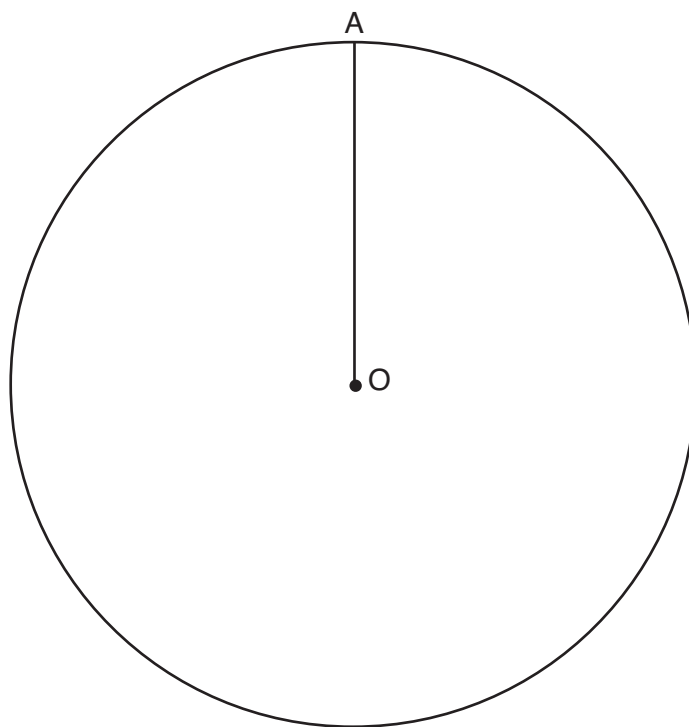
Describe a sequence of rigid motions that maps $\triangle ABC$ onto $\triangle STU$.

- 30** In right triangle PRT , $m\angle P = 90^\circ$, altitude $\overline{PQ}$ is drawn to hypotenuse $\overline{RT}$, $RT = 17$, and $PR = 15$.



Determine and state, to the *nearest tenth*, the length of $\overline{RQ}$.

- 31** Given circle O with radius $\overline{OA}$, use a compass and straightedge to construct an equilateral triangle inscribed in circle O . [Leave all construction marks.]



Part III

Answer all 3 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [12]

32 Riley plotted $A(-1,6)$, $B(3,8)$, $C(6,-1)$, and $D(1,0)$ to form a quadrilateral.

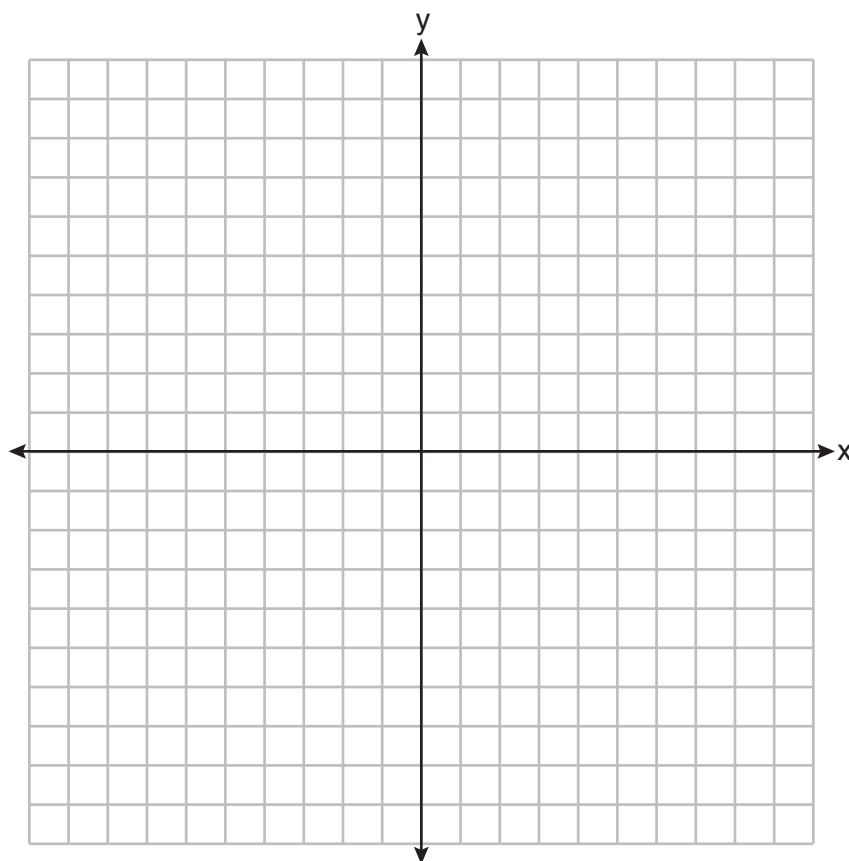
Prove that Riley's quadrilateral $ABCD$ is a trapezoid.

[The use of the set of axes on the next page is optional.]

Question 32 is continued on the next page.

Question 32 continued

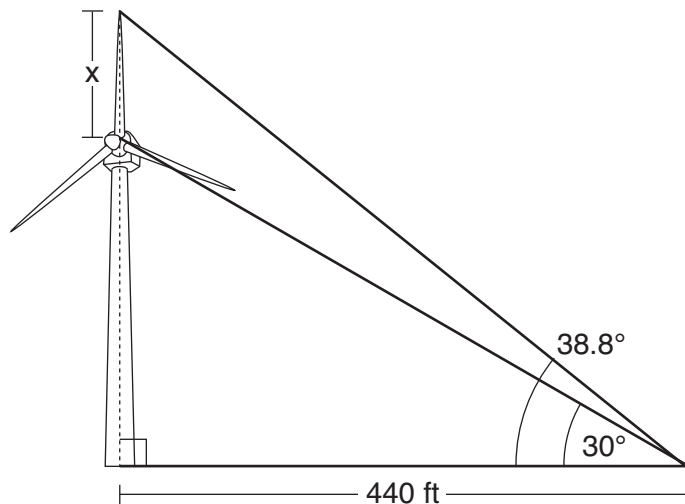
Riley defines an isosceles trapezoid as a trapezoid with congruent diagonals. Use Riley's definition to prove that $ABCD$ is *not* an isosceles trapezoid.



- 33** A child-sized swimming pool can be modeled by a cylinder. The pool has a diameter of $6\frac{1}{2}$ feet and a height of 12 inches. The pool is filled with water to $\frac{2}{3}$ of its height. Determine and state the volume of the water in the pool, to the *nearest cubic foot*.

One cubic foot equals 7.48 gallons of water. Determine and state, to the *nearest gallon*, the number of gallons of water in the pool.

- 34** Nick wanted to determine the length of one blade of the windmill pictured below. He stood at a point on the ground 440 feet from the windmill's base. Using surveyor's tools, Nick measured the angle between the ground and the highest point reached by the top blade and found it was 38.8° . He also measured the angle between the ground and the lowest point of the top blade, and found it was 30° .

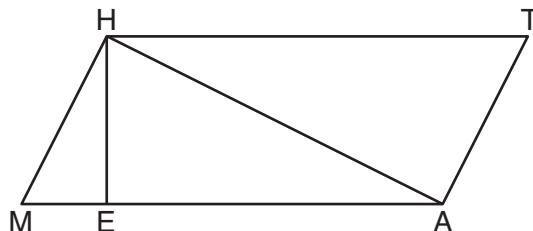


Determine and state a blade's length, x , to the *nearest foot*.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for the question to determine your answer. Note that diagrams are not necessarily drawn to scale. For the question in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

35 Given: Quadrilateral $MATH$, $\overline{HM} \cong \overline{AT}$, $\overline{HT} \cong \overline{AM}$, $\overline{HE} \perp \overline{MEA}$, and $\overline{HA} \perp \overline{AT}$



Prove: $TA \cdot HA = HE \cdot TH$

Work space for question 35 is continued on the next page.

Regents Examination in Geometry – June 2019**Scoring Key: Part I (Multiple-Choice Questions)**

Examination	Date	Question Number	Scoring Key	Question Type	Credit	Weight
Geometry	June '19	1	4	MC	2	1
Geometry	June '19	2	3	MC	2	1
Geometry	June '19	3	2	MC	2	1
Geometry	June '19	4	4	MC	2	1
Geometry	June '19	5	1	MC	2	1
Geometry	June '19	6	2	MC	2	1
Geometry	June '19	7	2	MC	2	1
Geometry	June '19	8	4	MC	2	1
Geometry	June '19	9	2	MC	2	1
Geometry	June '19	10	1	MC	2	1
Geometry	June '19	11	1	MC	2	1
Geometry	June '19	12	3	MC	2	1
Geometry	June '19	13	3	MC	2	1
Geometry	June '19	14	4	MC	2	1
Geometry	June '19	15	2	MC	2	1
Geometry	June '19	16	2	MC	2	1
Geometry	June '19	17	2	MC	2	1
Geometry	June '19	18	1	MC	2	1
Geometry	June '19	19	4	MC	2	1
Geometry	June '19	20	4	MC	2	1
Geometry	June '19	21	2	MC	2	1
Geometry	June '19	22	4	MC	2	1
Geometry	June '19	23	3	MC	2	1
Geometry	June '19	24	3	MC	2	1

Regents Examination in Geometry – June 2019**Scoring Key: Parts II, III, and IV (Constructed-Response Questions)**

Examination	Date	Question Number	Scoring Key	Question Type	Credit	Weight
Geometry	June '19	25	-	CR	2	1
Geometry	June '19	26	-	CR	2	1
Geometry	June '19	27	-	CR	2	1
Geometry	June '19	28	-	CR	2	1
Geometry	June '19	29	-	CR	2	1
Geometry	June '19	30	-	CR	2	1
Geometry	June '19	31	-	CR	2	1
Geometry	June '19	32	-	CR	4	1
Geometry	June '19	33	-	CR	4	1
Geometry	June '19	34	-	CR	4	1
Geometry	June '19	35	-	CR	6	1

Key

MC = Multiple-choice question

CR = Constructed-response question

The chart for determining students' final examination scores for the **June 2019 Regents Examination in Geometry** will be posted on the Department's web site at: <http://www.p12.nysed.gov/assessment/> on the day of the examination. Conversion charts provided for the previous administrations of the Regents Examination in Geometry must NOT be used to determine students' final scores for this administration.

**Map to the Learning Standards
Geometry
June 2019**

Question	Type	Credits	Cluster
1	Multiple Choice	2	G-CO.A
2	Multiple Choice	2	G-MG.A
3	Multiple Choice	2	G-GMD.B
4	Multiple Choice	2	G-CO.A
5	Multiple Choice	2	G-SRT.A
6	Multiple Choice	2	G-MG.A
7	Multiple Choice	2	G-SRT.A
8	Multiple Choice	2	G-CO.C
9	Multiple Choice	2	G-SRT.C
10	Multiple Choice	2	G-GMD.A
11	Multiple Choice	2	G-SRT.B
12	Multiple Choice	2	G-CO.C
13	Multiple Choice	2	G-C.A
14	Multiple Choice	2	G-CO.B
15	Multiple Choice	2	G-SRT.B
16	Multiple Choice	2	G-GPE.B
17	Multiple Choice	2	G-CO.C
18	Multiple Choice	2	G-C.A
19	Multiple Choice	2	G-GPE.B
20	Multiple Choice	2	G-GPE.A
21	Multiple Choice	2	G-CO.C
22	Multiple Choice	2	G-SRT.C
23	Multiple Choice	2	G-CO.C
24	Multiple Choice	2	G-CO.C
25	Constructed Response	2	G-CO.B
26	Constructed Response	2	G-GPE.B
27	Constructed Response	2	G-SRT.C
28	Constructed Response	2	G-C.B
29	Constructed Response	2	G-CO.B
30	Constructed Response	2	G-SRT.B
31	Constructed Response	2	G-CO.D
32	Constructed Response	4	G-GPE.B
33	Constructed Response	4	G-MG.A
34	Constructed Response	4	G-SRT.C
35	Constructed Response	6	G-SRT.B

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] No is indicated, and a correct explanation is written.

[1] No, but the explanation is incomplete or partially correct.

[0] No, but the explanation is missing or incorrect.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] 25, and correct work is shown.

[1] Appropriate work is shown, but one computational or graphing error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 25, but no work is shown.

[0] Triangle PQR is graphed correctly, but no further correct work is shown.

or

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) [2] 27, and correct work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] A correct relevant trigonometric equation is written, but no further correct work is shown.

or

[1] 27, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (28) [2] 20π , and correct work is shown.
- [1] Appropriate work is shown, but one computational error is made.
- or***
- [1] Appropriate work is shown, but one conceptual error is made.
- or***
- [1] Appropriate work is shown, but the answer is not in terms of π .
- or***
- [1] 20π , but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (29) [2] A correct sequence of rigid motions is written.
- [1] An appropriate sequence of rigid motions is written, but one computational error is made.
- or***
- [1] An appropriate sequence of rigid motions is written, but one conceptual error is made.
- or***
- [1] An appropriate sequence of rigid motions is written, but it is incomplete or partially correct.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (30) [2] 13.2, and correct work is shown.
- [1] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] A correct equation is written to find RQ , but no further correct work is shown.
- or**
- [1] 13.2, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (31) [2] A correct construction is drawn showing all appropriate arcs, and the equilateral triangle is drawn.
- [1] All appropriate arcs are drawn, but the triangle is not drawn.
- or**
- [1] Appropriate work is shown, but one construction error is made.
- or**
- [1] A correct construction of an equilateral triangle is drawn in circle O using the length of the radius as a side, but it is not inscribed in circle O .
- [0] A drawing that is not an appropriate construction is shown.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(32) [4] Correct work is shown to prove $ABCD$ is a trapezoid and not an isosceles trapezoid. Correct concluding statements are made.

[3] Appropriate work is shown, but one computational or graphing error is made. Concluding statements are written.

or

[3] Appropriate work is shown, but one concluding statement is missing or incorrect.

[2] Appropriate work is shown, but two or more computational or graphing errors are made.

or

[2] Correct work is shown to prove $ABCD$ is a trapezoid, and a correct concluding statement is written, but no further correct work is shown.

or

[2] Appropriate work is shown to prove $ABCD$ is not an isosceles trapezoid, and an appropriate concluding statement is written, but no further correct work is shown.

[1] Appropriate work is shown, but two or more computational or graphing errors are made, and one concluding statement is missing or incorrect.

or

[1] Appropriate work is shown to prove $ABCD$ is a trapezoid, but the concluding statement is missing or incorrect.

or

[1] Appropriate work is shown to prove $ABCD$ is not an isosceles trapezoid, but a method other than congruent diagonals is used.

or

[1] Appropriate work is shown to find the slopes of $\overline{AD}$ and $\overline{BC}$, but no further correct work is shown.

or

[1] Appropriate work is shown to find AC and BD , but no further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(33) [4] 22 and 165, and correct work is shown.

[3] Appropriate work is shown, but one computational or rounding error is made.

[2] Appropriate work is shown, but two or more computational or rounding errors are made.

or

[2] Correct work is shown to find 22, but no further correct work is shown.

[1] Appropriate work is shown to find the volume of the entire pool in cubic inches or cubic feet, but no further correct work is shown.

or

[1] 22 and 165, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (34) [4] 100, and correct work is shown.
- [3] Appropriate work is shown, but one computational error is made.
- or***
- [3] Appropriate work is shown to find the height to the bottom of the top blade and the height to the top of the top blade, but no further correct work is shown.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown to find the height to the bottom of the top blade or the height to the top of the top blade, but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational or rounding error are made.
- or***
- [1] At least one correct relevant trigonometric equation is written, but no further correct work is shown.
- or***
- [1] 100, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (35) [6] A complete and correct proof that includes a conclusion is written.
- [5] A proof is written that demonstrates a thorough understanding of the method of proof and contains no conceptual errors, but one statement and/or reason is missing or incorrect.
- [4] A proof is written that demonstrates a good understanding of the method of proof and contains no conceptual errors, but two statements and/or reasons are missing or incorrect.
- or**
- [4] A proof is written that demonstrates a good understanding of the method of proof, but one conceptual error is made.
- or**
- [4] Correct work is shown to prove $\triangle HEA \sim \triangle TAH$, but no further correct work is shown.
- [3] A proof is written that demonstrates a method of proof, but three statements and/or reasons are missing or incorrect.
- or**
- [3] A proof is written that demonstrates a method of proof, but one conceptual error is made and one statement and/or reason is missing or incorrect.
- [2] Some correct relevant statements about the proof are made, but four statements and/or reasons are missing or incorrect.
- or**
- [2] Correct work is shown to prove $\angle HEA \cong \angle TAH$ or $\angle EAH \cong \angle AHT$, but no further correct work is shown.
- [1] Only one correct statement and reason are written.
- [0] The “given” and/or the “prove” statements are written, but no further correct relevant statements are written.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
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